

Section 1: Foundations (State, Logic, & Types)

❖ Task: The E-Commerce Order Validator

❖ Scenario:

You are building the backend validation logic for an e-commerce store. You receive raw order data (which might be messy or null) and need to categorize the order status using modern Dart patterns.

❖ Requirements:

1. **Null Safety:** Create a nullable variable for couponCode (String?). Use the ?? operator to set a default "NO_COUPON" if it is null.
2. **Switch Expression:** Create a variable shippingCost. Use a **Switch Expression** on the orderTotal (double) to calculate cost:
 - Free if total > \$50.
 - \$5.00 if total is between \$20 and \$50.
 - \$10.00 otherwise.
3. **Pattern Matching:** Use a switch statement with **Guard Clauses (when)** to evaluate a "Status Tuple" (String status, bool isVip).
 - If status is "Pending" and user is VIP -> Print "Priority Processing".
 - If status is "Pending" -> Print "Standard Queue".
 - Default -> Print "Check Status".